Analytical Chemistry
Cumulative Exam
October 29, 2010

Theme: Problem Solving (Unannounced)

Instructions: Answer both questions as completely as you can. The goal of this cumulative exam is to determine your ability to solve chemical analysis problems with the cumulative knowledge in analytical chemistry that you have gained thus far in your graduate career. I am looking for thoughtful and well-reasoned ways to address the chemical analysis problem. There may be more than one "correct" answer. Be sure to completely comment on why you have chosen a particular instrumental method for the specified problem, instrumental limitations, sampling needs, and statistical analyses necessary for each method.

The questions below are derived from published papers in the last 6 months in the journal *Analytical Chemistry*. (This does not mean you have to solve it the way the authors reported.)

1) Halocarbons released into the atmosphere pose a potential concern. Devise a method (or methods) that could be used to detect the following halocarbons from the atmosphere: CH_3F (HFC-32), CF_3Br(Halon), CF_3I (Freon I3T1), CH_3CF_2Cl. Be sure to completely justify your answer with respect to all steps that would be necessary for your chosen analysis.

2) Classifying wine is important particularly with respect to identifying fakes. Consider a series of Chilean red wines (e.g., Syrah, Cabernet, and Merlot) that are grown in different vineyards. You wish to perform a complete metabolomic analysis for these wines. That means, you want to identify and quantify every small molecule (<1000 Da) component in the wine (e.g., glucose, other sugars, ethanol, bioflavonoids, phenols, etc.).

How could you perform such an analysis? What are the analysis challenges with this analysis?

Be sure to indicate what you could do to classify the wines in terms of the grapes, the location of the vineyard, etc. based on your chemical analysis. In other words, what types of statistical methods could be used to determine if a wine is a fake based on its chemical analysis. Justify your answer.